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Machine Learning Checklist v 1.1

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended to provide structure for consistency and transparency in reporting of works using or developing Machine Learning models. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

1. Availability and reproducibility of Code and Data

Please select all that apply regarding the availability of the data and code used in the study.

- ☐ Code will be included in a CodeOcean capsule.
- ☒ The **source code** is included in the submission or available in a public repository:
Prepared source: [Dendritic_credit_assignment_software.zip](#); upload pending.
- ☐ A **compiled standalone version** of the software is included in the submission or available in a public repository:
Not applicable: research source code, no standalone compiled application.
- ☒ A **test dataset** and instructions/scripts for replicating the results are included in the submission or available in a public repository:
Synthetic generators, test scripts and README in prepared reviewer archive.
- ☒ A **Readme file** with instructions for installing and running the code is included in the submission or available in a public repository:
Installation/run instructions in reviewer archive README and experiment READMEs.
- ☐ The code is made available to reviewers during review.
- ☐ **Pretrained models** are used in the study and accessible through:
Not applicable: no externally pretrained models.
- ☐ **Pretrained models** are used in the study and are not accessible.
- ☐ The paper contains information on how to obtain code and data after publication.

2. Datasets

- A. All data sources are listed in the paper.
 - ☒ Yes
 - ☐ No
- B. The train, test and validation datasets are publicly available, and links/accession numbers have been provided in the manuscript or supplementary materials.
 - ☐ Yes
 - ☐ No

- C. We have reported and discussed potential dataset biases in the paper. Where applicable, appropriate mitigation strategies were used.
- ☒ Yes Methods; SI S4-S10; Figs. S33-S52: cohort, finite-label, fixed-template and model-class limits.
- ☐ No _____
- D. The data cleaning and preprocessing steps are clearly and fully described, either in text or as a code pipeline.
- ☒ Yes Methods and SI S5-S10; executable preprocessing and response-baseline audit.
- ☐ No _____
- E. Instances of combining data from multiple sources are clearly identified, and potential issues mitigated.
- ☒ Yes Methods: anatomy/response joins; SI S6, S8 and S10; nested target/scan units.
- ☐ No _____

3. Model and training

- A. What model architecture is the current model based on? Conductance, linear, multi-affine trees (SI S1/S4/S5)
- B. A Model Card is provided¹.
- ☐ Yes
- ☒ No
- C. The model clearly splits data into different sets for training (model selection), validation (hyperparameter optimization), and testing (final evaluation).
- ☒ Yes
- ☐ No
- D. The method of data splitting (e.g. random, cluster- or time-based splitting, forward cross-validation) is clearly stated.
- ☒ Yes Methods; SI S4/S5/S10: separate task seeds, calibration and stimulus identities.
- ☐ No _____
- E. The data splitting mimics anticipated real-world applications.
- ☐ Yes _____
- ☒ No Mechanistic research: rotations; randomized interactions; seven fixed Boolean templates; observed-input visual responses.
- F. The data splitting procedure has been chosen to avoid data leakage.
- ☒ Yes SI S4: sealed selections; SI S10: nested training-only tuning and leakage audit.
- ☐ No _____

¹ <https://huggingface.co/docs/hub/model-cards>

G. The interpretability of the model has been studied and clearly validated.

- ☒ Yes Analytic gradients, explicit route dictionaries and controlled ablations; SI S1-S5.
- ☐ No _____

4. Evaluation

A. The performance metrics used are described and justified in the paper.

- ☒ Yes Methods and SI: accuracy, loss/NMSE, capture, update match, costed regret, selectivity.
- ☐ No _____

B. Cross-validation of the results is included.

- ☒ Yes
- ☐ No

C. Community-accepted benchmark datasets/tasks are used for comparisons.

- ☒ Yes MNIST, Fashion-MNIST and CIFAR-10; Methods and SI S5.
- ☐ No _____

D. Baseline comparisons to simple/trivial models (for example, 1-nearest neighbour, random forest, most frequent class) are provided.

- ☒ Yes Figs. 1-9/S31/S34-S52: broadcast, rank, decoder-only, point/ridge and privileged target-informed references.
- ☐ No _____

E. Benchmarks with current state-of-the-art are provided.

- ☐ Yes _____
- ☒ No No state-of-the-art accuracy or deployment claim; comparisons isolate mechanisms, information, optimization and resource constraints.

F. Ablation experiments are included.

- ☒ Yes Feedback/topology/optimizer/shunt; input grouping, exact/broadcast credit; cue and mapping controls.
- ☐ No _____

G. The model has been tested on a fully independent dataset.

- ☐ Yes
- ☐ No

5. Computational resources

A. The paper contains information on hardware/computing resources that were used.

- ☒ Yes
- ☐ No

B. The paper includes information on the computational costs in terms of computation time, parallelization or carbon footprints estimates.

- ☒ Yes
- ☐ No